Nishal Stanislaus Silva, *PhD*

Research Scientist | Real-Time Multimodal ML, Benchmarks & Open Datasets

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Research scientist (PhD, Trento) whose core discipline is benchmark and evaluation rigor: frozen-protocol test suites, controlled ablations, and a written record of negative results built to withstand scrutiny before a system is trusted. Domain expertise is real-time audio and multimodal time-series ML, not medical imaging or radiomics – that specific domain knowledge would need to be built. What transfers directly is the evaluation methodology itself (quantifying accuracy/latency/robustness trade-offs against a fixed protocol, exactly the discipline required to validate an imaging biomarker against clinical outcomes), hands-on VAE/diffusion work for limited-label regimes (medical imaging's chronic small-labeled-dataset problem), and one direct clinical deployment: a computer-vision device that digitized patient vitals across hospital wards during COVID-19, recognized by Sri Lanka's national physicians' association.

Core Competencies

Evaluation & Benchmark Rigor: Frozen evaluation protocols, controlled ablation design, statistical performance analysis, cross-method benchmarking (learned vs. classical baselines), negative-result documentation, reproducible dataset design.
Small-Labeled-Data Methods: VAE-based and diffusion-based synthetic time-series generation for limited-label regimes, training-free / low-data detection methods, transfer and fusion across heterogeneous sensor modalities.
Human-Centred Evaluation: Quantitative + qualitative user studies, structured interviews, real-world usability and adoption assessment (audience/performer studies, clinical-deployment feedback).

Technical Stack

Python (TensorFlow, Keras, PyTorch, scikit-learn, SciPy, NumPy, Pandas), C++17, Computer Vision (OpenCV), SQL, AWS, Docker, ETL pipelines, CI/CD, Git, statistical analysis and experiment design, LaTeX.

Education

PhD, Information Engineering and Computer Science – University of Trento, Italy (2021 - 2025)
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK (2016 - 2018)
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK (2013 - 2014)

Research and Work Experience

Independent Researcher | *Self-directed, Toronto, Canada* *Jan 2026 - Present*

- Continued peer-reviewed research: two papers accepted (Smart Drums, JAES; MIRaaS, IEEE IS2) covering real-time pattern detection and multimodal information retrieval.
- Maintain Nebula, an open-source C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks.
- Applied AI-assisted engineering workflows, including Claude Code, to accelerate research tooling, documentation, and pipeline development.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio and IMU/gesture data on the EU Horizon EIC Pathfinder project MUSMET: acquisition, DSP/feature engineering, training and evaluation, edge deployment, and monitoring.

- Designed frozen-protocol benchmark suites and controlled ablations (RNN vs. DTW; audio vs. MIDI; pattern-set sizes 1/3/10) that quantified accuracy, latency, and CPU trade-offs, with a written record of negative results.
- Published a real-time detector achieving 14 ms inference on Raspberry Pi 4, F1 0.76, 31.4% CPU utilization – 74x faster than the 1038 ms DTW baseline it replaced (JAES 2026).
- Prototyped VAE-based and diffusion-based synthetic time-series generation to address limited-label regimes.

Visiting Researcher | *McGill University, IDMIL/CIRMMT, Montreal, Canada* *May 2024 - Aug 2024*

- Built a self-powered smart electric guitar (embedded IMU and audio capture); fused IMU and audio streams via a hierarchical finite-state-machine classifier, achieving F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka* *Jan 2024 - Mar 2024*

- Ran a human-centred evaluation study of musicians' perception of smart-instrument pattern detection, combining quantitative metrics with structured interviews to assess real-world usability and adoption (IEEE I3DA 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Built ML and statistical pipelines for financial forensics, compliance monitoring, and risk detection on large structured enterprise datasets.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* *Jan 2015 - Jul 2020*

- 5.5 years delivering end-to-end computer-vision and IoT systems for manufacturing at scale, driving operational efficiency gains of up to 300%.
- Developed and deployed a computer-vision device for remote monitoring of patient vitals across hospital wards during COVID-19, recognized by the Government Medical Officers' Association (GMOA), Sri Lanka – direct experience shipping a clinical-facing CV system with real healthcare-worker adoption stakes.
- Automated multilingual care-label quality control (OpenCV template and feature matching) integrated with conveyor and PLC reject logic, cutting inspection time by 99.5%.
- Built real-time IoT data pipelines with C++/Python inference on embedded hardware; introduced CI/CD and code review practices; mentored engineers.

Publications

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *J. Audio Eng. Soc.* 74(3). Silva, N. & Turchet, L. (2026, in press). Smart Drums. *J. Audio Eng. Soc.* Silva, N. & Turchet, L. (2026, accepted). MIRaaS. *IEEE IS2*. Silva, N., Wanderley, M. & Turchet, L. (2025). Melody and Motion. *IEEE IS2*. Silva, N., Boem, A. & Turchet, L. (2025). Interactive IoMusT-Based Concerts. *IEEE I3DA* – a human-study paper with real audience and performer evaluation. Silva, N. & Turchet, L. (2022). An SSIM-based training-free onset-detection method achieving 95% detection with no training data. *DAFx*.

Datasets

Creator of four open-access musical-pattern datasets on Zenodo (DoMP, DoPP, DoDP, DoDP2), comprising 9,800+ recordings from 70+ musicians, each released with a frozen evaluation protocol supporting reproducible benchmarking.

Highlights

MIDI Innovation Awards 2023 Finalist (Prototypes & Non-Commercial Software category). GMOA recognition (Government Medical Officers' Association, Sri Lanka, 2020) for the COVID-19 remote-vitals monitoring deployment. Peer reviewer for IEEE Access and JNSF Sri Lanka; programme committee member for IEEE IS2 and IEEE I3DA; supervised two BSc students.